

Algebraic Geometry 1

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Abstract

TODO

Chapter 1

Affine varieties

Lecture 1: Syllabus day

Thu 03 Sep 2026 11:07

We will follow Andreas Gathmann's 2002 notes [Gat02], and will supplement them with the various books on Algebraic Geometry, such as [Vak24] and Hartshorne.

Algebraic geometry is essentially the study of the geometry of solution sets to systems of polynomial equations, using the tools of commutative algebra. We set up a dictionary to translate between geometric and algebraic properties, which holds more generally than just polynomial rings. Given a sufficiently nice space X , we obtain a ring $C(X)$ of $\mathbb{C}$ -valued continuous functions. There's a number of things we can do here. For example, given a point $x \in X$, there is an evaluation map $\varepsilon_x : C(X) \rightarrow \mathbb{C}$ which ends up being a ring homomorphism. Since the space is nice, we can take ε_x to be surjective. Thus, $\text{Ker } \varepsilon_x$ is a maximal ideal of $C(X)$ due to the first isomorphism theorem $C(X)/\text{Ker } \varepsilon_x \cong \mathbb{C}$. Thus, something as simple as a point $x \in X$ manifests as a maximal ideal in the ring $C(X)$.

More generally, let $Z \subseteq X$ be closed. I can define an ideal $I_Z \subseteq C(X)$ of all functions $f \in C(X)$ that vanish on Z . Thus, arbitrary closed sets give us an ideal that is not necessarily maximal.

Consider the space X given by the union of the x - and y -axes inside of $\mathbb{C}^2$. Letting $f = y$ and $g = x$, we see that f vanishes on the y -axis, and g vanishes on the x -axis. Then $fg = xy$ is identically zero on X . Thus, the ring $C(X)$ has zero divisors.

1.1 Algebraic sets and the Zariski topology

Definition 1.1.1. Let k be a field. Define the n -dimensional affine space over k , denoted $\mathbb{A}_k^n$, as

$$\mathbb{A}_k^n := \{(x_1, \dots, x_n) \mid x_i \in k \forall i\}.$$

The intuition for affine space is that we try to think of it as an n -dimensional vector space without an origin. We will also want to consider non-linear systems of equations over affine space, which will lead us to topologize it in a strange way. This topology is what differentiates $\mathbb{A}_k^n$ from k^n in the case where k is a topological field.

For now, note that we can view $k[x_1, \dots, x_n]$ as k -valued polynomial functions on $\mathbb{A}_k^n$. Given a subset $S \subseteq k[x_1, \dots, x_n]$, we want to find the *solution set* or *vanishing locus*

$$Z(S) = \{x \in \mathbb{A}_k^n \mid \forall p \in S, p(x) = 0\}.$$

We can make some observations. If $S_1 \subseteq S_2 \subseteq k[x_1, \dots, x_n]$, then since S_2 applies more relations, $Z(S_2) \subseteq Z(S_1)$. Thus Z is inclusion reversing. The next observation is that it would be more natural

to consider ideals. This is further motivated by the fact that if (S) is the ideal generated by S , then we can show $Z(S) = Z((S))$. To prove this, first note that since $S \subseteq (S)$, we have $Z((S)) \subseteq Z(S)$. For the converse, let $x \in Z(S)$. Note that an element of (S) is of the form

$$\sum_i g_i p_i, \quad g_i \in k[x_1, \dots, x_n], \quad p_i \in S.$$

Since $p_i(x) = 0$ for all i , this sum must vanish at x . If $S = \{f_1, \dots, f_r\}$ is finite, then we often write $Z(f_1, \dots, f_r)$. This discussion leads us to give sets of the form $Z(S)$ a special name.

Definition 1.1.2. An *algebraic set* is a subset of $\mathbb{A}_k^n$ of the form $Z(S)$, where $S \subseteq k[x_1, \dots, x_n]$.

For example, $\mathbb{A}_k^n = Z(0)$, so affine space is an algebraic set. The empty set is the vanishing locus of any nonzero constant function, so it is also an algebraic set. A single point $x \in \mathbb{A}_k^n$ is the vanishing locus of $Z(x_1 - a_1, \dots, x_n - a_n)$, so singletons are algebraic sets. More generally, any linear or affine linear subspace of $\mathbb{A}_k^n$ is an algebraic set. For example, $x_2 - x_1$ cuts out a diagonal in the $x_1 x_2$ -plane. The circle is the vanishing locus of $(x_1^2 + x_2^2 - 1)$.

We will now define the topology we alluded to earlier, referred to as the *Zariski topology*. The idea is that we want to topologize $\mathbb{A}_k^n$ such that the closed sets are precisely the algebraic sets.

Proposition 1.1.3. *There is a topology on $\mathbb{A}_k^n$ whose closed sets are precisely the algebraic sets.*

Proof. We must show the following.

1. $\emptyset, \mathbb{A}_k^n$ are closed,
2. a finite union of closed sets is closed, and
3. arbitrary intersections of closed sets are closed.

We've already shown property 1. The remaining results follow immediately from the following lemma – since the relevant constructions end up being vanishing loci of ideals, the result follows. ■

Lemma 1.1.4. *Let $\{I_j\}_{j \in J}$ be ideals of $k[x_1, \dots, x_n]$. Then*

$$\bigcap_{j \in J} Z(I_j) = Z\left(\sum_{j \in J} I_j\right).$$

Additionally, if $J = [n]$ is finite, then

$$\bigcup_{j=0}^n Z(I_j) = Z\left(\prod_{j=0}^n I_j\right) = Z\left(\bigcap_{j=0}^n I_j\right).$$

Proof. Recall a sum of ideals is all finite linear combinations in those ideals, and the product of ideals is given by all monomials using the normal ring product (so thing $gf_1 \cdots f_n$). ■

This topology is different from the usual topology on $\mathbb{A}_k^n$ (when $k = \mathbb{R}, \mathbb{C}$). We call this the *Zariski topology*. Whenever we refer to affine space, we implicitly take the given topology to be the Zariski topology.

Let's look at some closed sets of $\mathbb{A}_k^1$. Let f be a polynomial over an algebraically closed field k . Then f splits as $(x - a_1) \cdots (x - a_r)$, so $Z(f) = \{a_1, \dots, a_r\}$. Thus since k is algebraically closed,

the closed sets of 1 dimensional affine space are finite collections of points. Thus the opens are FUCKING huge in the Zariski topology, and the Zariski topology is *far* coarser than the analytic topology. In fact, any two nonempty opens will intersect in $\mathbb{A}_k^1$ (and I think in any affine space). Of course, closed sets in the Zariski topology can have uncountably many points in higher dimensions (recall all our earlier examples).

Finally, we recall Hilbert's basis theorem, which will be useful later on in the course.

Theorem 1.1.5 (Hilbert's basis theorem). *Let R be a Noetherian ring. Then $R[x]$ is a Noetherian ring.*

Lecture 2: Affine varieties

Tue 08 Sep 2026 11:02

We now have a course page [here](#). We will now be following Galthmann's latest notes [Gat21]. Note that these notes define an affine variety as what we previously called an algebraic set.

Definition 1.1.6 (Affine variety). An *affine variety* is an algebraic set (see definition 1.1.2). Some authors (and our original notes) require that affine varieties be *irreducible* algebraic sets, but we do not make that distinction. Thus, we refer only to affine varieties in the sequel.

Let k be a field, usually algebraically closed. We will switch to denoting the vanishing locus of $I \subseteq k[x_1, \dots, x_n]$ by $V(I)$ rather than $Z(I)$. Recall that we topologize affine space $\mathbb{A}_n^k = k^n$ in a nonstandard way, by taking the closed sets to be precisely $Z(I)$ for all ideals I (equivalently subsets $S \subseteq k[x_1, \dots, x_n]$, since $V(S) = V((S))$, for (S) the ideal generated by S).

Exercise 1.1.7. Consider $\mathbb{A}_k^3$. Let I be the ideal $(x^2 + y^2 + z^2 - 1)$. Let J be the ideal (x, y) . What is $V(I) \cap V(J)$?

Proof. We expect $V(I)$ to be the sphere, and we expect J to be the z -axis. Thus the intersection should be the two intersection points $(0, 0, \pm 1)$. By lemma 1.1.4, we have $V(I) \cap V(J) = V(I + J)$. We then have

$$V(I + J) = V(x, y, x^2 + y^2 + z^2 - 1).$$

Thus we need a point $(x, y, z) = (0, 0, z)$ with $z^2 - 1 = 0$. This indeed agrees with our geometric intuition (since the field is algebraically closed, yielding 2 solutions). ■

Recall that if R is a commutative ring and $I \subseteq R$ is an ideal, then the radical $\sqrt{I}$ is the ideal of all n th roots of elements of I , defined more precisely as

$$\sqrt{I} = \{x \in R \mid x^n \in I\}.$$

Definition 1.1.8 (Radical). Let R be a commutative ring, and let $I \subseteq R$ be an ideal. The *radical ideal* of I , denoted $\sqrt{I}$, consists of all n th roots of elements of I . More precisely,

$$\sqrt{I} := \{x \in R \mid \exists n \in \mathbb{N} \text{ s.t. } x^n \in I\}.$$

Exercise 1.1.9. Show that

$$\sqrt{I} = \bigcap_{\substack{\mathfrak{p} \text{ prime} \\ I \subseteq \mathfrak{p}}} \mathfrak{p}.$$

If $I = 0$, then $\sqrt{I} = \sqrt{0}$ is denoted $\mathfrak{N}$, and called the *nilradical*. It consists of all nilpotent elements of R , and by ?? it follows that $\mathfrak{N}$ is the intersection of all prime ideals of R .

Fun fact: if $\pi : R \rightarrow R/I$ is the quotient map, then $\sqrt{I} = \pi^{-1}(\mathfrak{N})$. Another one: $\sqrt{(x^n)} = (x)$. We also have $\sqrt{(x^2, y^3)} = (x, y)$.

Let j be the map taking ideals in $k[x_1, \dots, x_n]$ to closed subsets in $\mathbb{A}_k^n$ by mapping $I \mapsto V(I)$. We claim that $j(I) = j(\sqrt{I})$. For example, $V(x^4) = 0 = V(x)$. We can consider the inverse direction. Let $X \subseteq \mathbb{A}_k^n$; we can look at the ideal $I(X)$ of all functions $f \in k[x_1, \dots, x_n]$ that vanish on X . This yields maps

$$j : \{\text{ideals of } k[x_1, \dots, x_n]\} \rightleftarrows \{\text{closed subsets of } \mathbb{A}_k^n\} : I.$$

How do these maps relate? In other words, what are $I(j(J)) = I(V(J))$ and $V(I(X))$ for J an ideal and X a closed subset? This follows by Hilbert's nullstellensatz.

Theorem 1.1.10 (Hilbert's nullstellensatz). *Let $X \subseteq \mathbb{A}_k^n$ be a closed subset, and let k be an algebraically closed field.*

1. $V(\sqrt{I(X)}) = V(I(X)) = X$.
2. $I(X) = \sqrt{I(X)}$.

Consider an example. Let $J \subseteq k[x]$ to be the ideal $J = ((x - a_1)^{n_1} \cdots (x - a_r)^{n_r})$, for $n_j > 0$ and $a_i \neq a_j$ when $i \neq j$. Write f for this polynomial, such that $J = (f)$. Since $k[x]$ is a principal ideal, $\sqrt{J}$ is generated by a single element. We have

$$\sqrt{J} = ((x - a_1) \cdots (x - a_r)).$$

We see that the vanishing loci of J and $\sqrt{J}$ are clearly equal.

Remark 1.1.11. Recall by ?? that all radicals are intersections of primes. Thus, any prime ideal $\mathfrak{p}$ is radical. Moreover, all maximal ideals are radical. Given affine varieties $X_1, X_2 \subseteq \mathbb{A}_k^n$, we can even show

$$I(X_1 \cup X_2) = I(X_1) \cap I(X_2), \quad I(X_1 \cap X_2) = \sqrt{I(X_1) + I(X_2)}.$$

(exercise: how to prove this?) This implies that intersections of radicals are radical.

Lecture 3: Investigating affine space

Thu 10 Sep 2026 11:03

Note that the singletons of affine space are closed, and correspond precisely to maximal ideals.

Definition 1.1.12. Let $X \subseteq \mathbb{A}_k^n$ be an affine variety. Define the *coordinate ring* of X to be

$$\frac{k[x_1, \dots, x_n]}{I(X)}.$$

Note that since $I(X) = \sqrt{I(X)}$, the coordinate ring is *reduced*, meaning it has no nilpotent element.

let $X = V(\langle x_1 + x_2 + x_3 \rangle) \subseteq \mathbb{A}_k^3$ be a plane. Then $A(X) = k[x_1, x_2, x_3]/(x_1 + x_2 + x_3)$ is the coordinate ring. Pick the ideal $\bar{J} = \langle x_3 \rangle$. Then $\pi^{-1}(\bar{J}) = \langle x_3, x_1 + x_2 + x_3 \rangle$. Rewriting this as an intersection as varieties this is a line $x_1 + x_2$ in the $(x_3 = 0)$ -plane.

Let's return to our discussion of Zariski. Recall that we topologize $\mathbb{A}_k^n$ such that closed subsets are precisely the affine varieties $V(I)$. The subspace topology and the natural Zariski topology on a subvariety will coincide.

Lemma 1.1.13. *Let $X \subseteq \mathbb{A}_k^n$ and $Y \subseteq \mathbb{A}_k^m$ be affine varieties over the same field. Then the product $X \times Y \subseteq \mathbb{A}_k^n \times \mathbb{A}_k^m$ has the structure of an affine variety (as sets).*

To see why there are subtleties here, consider the product of

$$X = V(\langle x_2 - x_1^2 \rangle), \quad Y = V(\langle y_2^3 - y_1^2 \rangle).$$

Then let's consider

$$I(X \times Y) \subseteq k[x_1, x_2, y_1, y_2] \cong k[x_1, x_2] \otimes_k k[y_1, y_2].$$

We have

$$I(X \times Y) = \langle x_2 - x_1^2, y_2^3 - y_1^2 \rangle.$$

Proof. If $I(X) = \langle f_1(x_1, \dots, x_n), \dots, f_r(x_1, \dots, x_n) \rangle$ and $I(Y) = \langle g_i(y_1, \dots, y_m) \rangle_{1 \leq i \leq s}$, (here we use R is Noetherian) then

$$X \times Y = V(\langle f_1, \dots, f_r, g_1, \dots, g_s \rangle) \subseteq k[x_1, \dots, y_m].$$

Note:

$$\langle f_1, \dots, g_s \rangle = I(X) \otimes 1 + 1 \otimes I(Y) \subseteq k[x_1, \dots, x_n] \otimes_k k[y_1, \dots, y_m].$$

■

Issue: the Zariski topology on $X \times Y$ is *not* the product topology. We will later rectify this issue.

Now consider irreducibility and irreducible components. For example, consider $V(\langle x_1, x_2 \rangle) \subseteq \mathbb{A}_k^2$. Note that this is just the axes, so we can write

$$V(\langle x_1, x_2 \rangle) \subseteq V(\langle x_1 \rangle) \cup V(\langle x_2 \rangle).$$

Thus, we can write our variety as a subset of two proper closed subvarieties. However, we cannot do this for the axes themselves: we cannot write them as a finite union of closed subsets. Here, recall that an axis is just $\mathbb{A}^1$ whose closed subsets are collections of points. Obviously a line is not a finite union of finite points.

Definition 1.1.14. Let X be a topological space. We say X is *reducible* if there are proper closed subsets $X_1, X_2 \subsetneq X$ such that $X_1 \cup X_2 = X$. Otherwise, X is *irreducible*. The space X is *disconnected* if those subsets X_1, X_2 can be taken to be disjoint, and otherwise X is *connected*.

Lecture 4: Sheaves, apparently

Tue 22 Sep 2026 11:02

I missed 2 classes and did not learn about dimension and some stuff about Zariski's topology. I will add those to my notes after this lecture.

Here's a recap of last time. Given a space X , define the dimension of X as

$$\dim X := \sup \{n \in \mathbb{N} \cup \{\infty\} \mid \exists \text{ a chain } \phi \subseteq Y_0 \subsetneq Y_1 \subsetneq \dots \subsetneq Y_n \subseteq X \text{ of irreducible closed subsets}\}.$$

If $Y \subseteq X$ is an irreducible closed subset, then its codimension $\text{codim}_X Y$ of Y in X is the sup over all chains that *start* at Y , so $Y = Y_0$.

When X is an affine variety, this notion of dimension coecides with the Krull dimension of $A(X)$. If R a commutative ring, then $\dim_{\text{Kr}}(R)$ is the sup over possibly infinite lengths n of chains of prime ideals $0 \subseteq \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n \subsetneq R$. The codimension of a prime ideal is defined analogously, with $\mathfrak{p} = \mathfrak{p}_n$ (The Nullstellensatz is inclusion reversing), and is called its *height*.

Here are some facts about dimension theory in commutative algebra.

Theorem 1.1.15. *Let X be an irreducible affine variety. Note that products are of varieties, not of spaces.*

1. $\dim_{\text{Kr}} A(X) = \dim X$ is finite.
2. If Y is another irreducible affine variety, then $X \times Y$ has dimension $\dim X + \dim Y$.
3. If $Y \subseteq X$ is irreducible, then $\dim X = \dim Y + \text{codim}_X Y$.
4. If $f \in A(X)$, then every irreducible component of $V(f)$ has codimension 1 in X . Hence we say $\dim V(f) = \dim X - 1$. (Krull's principle ideal theorem)

Note that $\dim \mathbb{A}^1 = 1$. This implies $\dim \mathbb{A}^2 = \dim(\mathbb{A}^1 \times \mathbb{A}^1) = 2$. Inductively, $\dim(\mathbb{A}^n) = n$.

Another corollary is that if $f \in k[x_1, \dots, x_n]$, then $\dim V(f) = n - 1$. We call $V(f)$ the *hypersurface corresponding to f* .

Chapter 2

The sheaf of regular functions

this should be chapter 3, fix later

We will now rigorize the intuition that elements of the coordinate ring can be thought of as k -valued polynomial functions on their affine variety.

Definition 2.0.1. Let X a space and $\mathcal{C}$ a category. A *presheaf* $\mathcal{F}$ on X valued in $\mathcal{C}$ is a functor $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$. If $U \subseteq V$, we call $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$ a restriction morphism. We say $\mathcal{F}$ is a *sheaf* if for all opens $U \subseteq X$ and all open covers $\{U_i\}_{i \in I}$ of U , the diagram

$$\mathcal{F}(U) \longrightarrow \prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i, j \in I} \mathcal{F}(U_i \cap U_j)$$

exists, and is an equalizer. Here the parallel arrows are given by the maps $U_i \cap U_j \subseteq U_i$ and $U_i \cap U_j \subseteq U_j$, i.e. inserting into the left and right sides.

If $\mathcal{C}$ is concrete in the sense that the forgetful functor U can be given as $U \cong \mathcal{C}(*, -)$ with $*$ terminal, then since Hom is continuous and covariant in its second argument, we have a nice characterization of this behavior. Elements of $\mathcal{F}(U)$ are called *sections*. If $s \in \mathcal{F}(V)$ is a section and $U \subseteq V$, then we write $s \mapsto s|_U$ for the restriction $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$. We now reformulate the equalizer as follows.

If $U \subseteq X$ is an open with $\{U_i\}_{i \in I}$ an open cover, and $\{s_i \in \mathcal{F}(U_i)\}_{i \in I}$ is a collection of sections such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there is a unique section $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all $i \in I$. Also since $\mathcal{F}$ can be computed as a limit, and $\emptyset$ is a colimit (initial) in $\text{Open}(X)$ (thus a limit in the opposite category), we see that $\mathcal{F}(\emptyset)$ is terminal in $\mathcal{C}$.

Examples.

1. Let M a manifold. Then the sheaf of rings C_M^∞ of smooth maps on M .
2. Let X a complex manifold. Then the sheaf of rings $\mathcal{O}_X$ of holomorphic maps $X \rightarrow \mathbb{C}$.
3. Let X a space. Then the sheaf of rings $C_X^0 = \mathcal{T}\text{op}(-, \mathbb{R})|_{\text{Open}(X)^{\text{op}}}$ of continuous maps $X \rightarrow \mathbb{R}$.
4. Let X a space and R a topological ring. Then the sheaf of rings $C_X^0(-, R) = \mathcal{T}\text{op}(-, R)|_{\text{Open}(X)^{\text{op}}}$ of continuous maps $X \rightarrow R$.
5. Let M a manifold, and $E \rightarrow M$ a smooth vector bundle. Then the sheaf of $\mathbb{R}$ -vector spaces $\Gamma(-, E) = \mathcal{E}$ of smooth sections of E . Similarly for continuous sections of a continuous bundle, and holomorphic sections of a holomorphic bundle.

6. Let M a manifold. Then the de Rham complex $\Omega_M^\bullet$ is a sheaf valued in cochains over $\mathbb{R}$.
7. Let M be a manifold with $E \rightarrow M$ a graded vector bundle. Then $\Gamma(-, E)$ is a sheaf of graded vector spaces. If there is a map $d : E \rightarrow E$ with $d^2 = 0$, then $\Gamma(-, E)$ takes values in cochains over $\mathbb{R}$.

Things that cannot be defined with a local property are usually not sheaves. For example, consider the sheaf L^2 on $\mathbb{R}$ of measurable functions with finite L^2 -norm. For example, consider the function $f(x) = x$. On some bounded $U \subseteq \mathbb{R}$, it has an integral. However, we can cover U by finite intervals on which $f(x)$ is square-integrable. But the only thing $f(x)$ glues to is $f(x) = x$, which has an infinite integral over all of $\mathbb{R}$. Hence, we cannot glue this section.

For another example, consider the complex manifold $\mathbb{C} \setminus \{0\} =: \mathbb{C}^*$. Define $\mathcal{F}$ to be the sheaf of holomorphic functions f such that $e^f = 1$. In other words, $\mathcal{F}$ is the sheaf of branches of the complex logarithm. We cannot define this globally, so $\mathcal{F}(\mathbb{C}^*)$ is empty. But we can cover $\mathbb{C}^*$ by opens that *have* a complex logarithm, hence there is no gluing.

Definition 2.0.2. Let X be a space, and $\mathcal{O}_X$ a sheaf of rings on X . We call $(X, \mathcal{O}_X)$ a *ringed space*, and call $\mathcal{O}_X$ the *structure sheaf* of X . We often will simply say “let X be a ringed space”, and leave the $\mathcal{O}_X$ implicit.

Now we will begin our discussion of the structure sheaf of an affine variety. Let X be an affine variety. We have the coordinate ring $A(X)$ of functions on the whole variety, but we also need to be able to consider local information. A good solution will be *regular* functions.

Definition 2.0.3. Let X be an affine variety over k , and $U \subseteq X$ an open set. Then a *regular function* on U is a map $\varphi : U \rightarrow k$ such that for every $a \in U$, there exists an open $U_a \subseteq U$ and polynomials $f, g \in A(X)$ such that $f(x) \neq 0$ for all $x \in U_a$, and $\varphi(x) = g(x)/f(x)$ for all $x \in U_a$. We denote the set of all regular functions on U by $\mathcal{O}_X(U)$.

For example, the function $\varphi(x) = p(x)/x^n$ is reasonable on $\mathbb{A}^1 \setminus \{0\} \subseteq \mathbb{A}^1$, where p is a polynomial on $\mathbb{A}^1$.

Lecture 5: Sheaves, more

Thu 24 Sep 2026 11:00

Here is another example of a regular function. The idea is to show that the expression $\varphi = g/f$ can vary in g, f depending on the location, hence we cannot always choose a global representation. Let $X = V(x_1x_4 - x_2x_3) \subseteq \mathbb{A}_k^4$. Given any solution x in the variety, and some $\lambda \in k$, it follows that λx also lies in this variety. We call such an object a *cone*. This follows since the polynomial is homogeneous of degree 2.

Consider the opens $U_1 = X \setminus V(x_2)$ and $U_2 = X \setminus V(x_4)$. Then if U is their union, then note that this is precisely $X \setminus V(x_2, x_4)$, since this is the locus where at least one does not vanish.

Define $\varphi : U \rightarrow k$ by

$$x \mapsto \begin{cases} \frac{x_1}{x_2}, & x \in U_1 \\ \frac{x_3}{x_4}, & x \in U_2 \end{cases}.$$

We claim this is well-defined. If $x \in U_1 \cap U_2$, then $\frac{x_1}{x_2} = \frac{x_3}{x_4}$ if and only if $x_1x_4 = x_2x_3$. This is precisely the defining equation of X , hence $x \in X$ implies that this is well-defined manifestly.

However, the top representation is *not valid* near the point $(0, 0, 0, 1)$, since $x_2 = 0$. Note that this point falls in U_2 , since x_4 is nonvanishing here. Conversely, $\frac{x_3}{x_4}$ is not valid near $(0, 1, 0, 0)$. Hence we see that φ has distinct representations near these points.

Proposition 2.0.4. *The sheaf of regular functions is a sheaf.*

Proof. Let X an affine variety, and $\mathcal{O}_X$ its presheaf of regular functions. Here we note that the fact this is a presheaf is obvious. The rest is tedious so we only sketch the proof. Note that $\mathcal{O}_X$ is a sheaf of rings, taking values in $A(X \text{ or } U??)[W]$ for some W that I can't figure out.

To show gluing, we use the local definition of regular functions. Let $\{\varphi_i\}_{i \in I}$ be regular functions on an open cover $\{U_i\}_{i \in I}$ of U . Define a section φ over U as follows. Given $x \in U$, we have $x \in U_i$ for some i . Define $\varphi(x) = \varphi_i(x)$. If φ_j is defined at x , then $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$ implies $\varphi_j(x)$. Since each of the φ_i 's is regular, this is locally given as a quotient of two polynomials. Hence, so is φ . ■

Remark 2.0.5. Note that we did not use the local definition of a regular function extensively. However, it is important that we had this. If we required that it had a global formula, then we might not be able to glue, since we would have to find a formula that all φ_i 's simultaneously satisfy. This is not possible in general.

Lemma 2.0.6. *Let $U \subseteq X$ be open, and $\varphi \in \mathcal{O}_X(U)$ a regular function. Then the vanishing locus $V(\varphi)$ is closed in U .*

Proof. For all $a \in U$, there is some $a \in U_a \subseteq U$ with $\varphi = g_a/f_a$, $g_a, f_a \in A(X)$. Ok I don't care i care more about the following ■

Claim 2.0.7. $\mathcal{O}_X(U)$ is a *fucking ring*.

Proof. Let $\varphi, \psi \in \mathcal{O}_X(U)$. Then for all $a \in U$, there is U_a and V_a with $a \in U_a, V_a \subseteq U$ and $\varphi = g/f$ on U_a , $\psi = k/h$ on V_a . We work in the intersection $U_a \cap V_a \ni a$, which is open. Then

$$(\varphi + \psi)(x) = \frac{g(x)}{f(x)} + \frac{k(x)}{h(x)} = \frac{g(x)h(x) + f(x)k(x)}{f(x)h(x)}.$$

Since $f(x), h(x) \neq 0$ on U_a and V_a respectively, the same holds on the intersection. Thus this is a well-defined function, so for all a , we have $a \in V_a \cap U_a \subseteq U$ with $\varphi + \psi$ equal to a quotient. For multiplication, it's much easier since we just have gk/fh . Additionally, since the constant function is a polynomial, this also shows $\mathcal{O}_X(U)$ is a k -algebra. ■

Proposition 2.0.8. *Let X be irreducible, and $U \subseteq V \subseteq X$ opens of an affine variety. If $\varphi_1, \varphi_2 \in \mathcal{O}_X(V)$ agree on U , then $\varphi_1 = \varphi_2$.*

Proof. Note that $V(\varphi_1 - \varphi_2)$ is closed, and we have $U \subseteq V(\varphi_1 - \varphi_2)$. Since X is irreducible, any open is dense. Thus U is dense in $V(\varphi_1 - \varphi_2)$, hence $\overline{U} = V(\varphi_1 - \varphi_2)$. However, U is also dense in X , hence V , so $\overline{U} = X = V$. Thus $V(\varphi_1 - \varphi_2) = V$, so we are done. ■

We now discuss the basis of distinguished open sets of the Zariski topology. For an arbitrary open $U \subseteq X$, it may be difficult to compute $\mathcal{O}_X(U)$. For distinguished opens, the ring is quite tractable as a localization.

Definition 2.0.9. Let $f \in A(X)$. Then the *distinguished open* $D(f)$ is the nonvanishing locus of f :

$$D(f) = X \setminus V(f) = \{x \in X \mid f(x) \neq 0\}.$$

This is clearly open.

Claim 2.0.10. *Let $f, g \in A(X)$. Then $D(f) \cap D(g)$ is a distinguished open.*

Proof.

$$D(f) \cap D(g) = D(fg).$$

■

Claim 2.0.11. *If $U \subseteq X$ is open, then there is an open cover of U by distinguished opens.*

Proof. Since U is open, we have $U = X \setminus V(J)$ for some ideal $J \subseteq A(X)$. Since $A(X)$ is Noetherian, we have $J = \langle f_1, \dots, f_r \rangle$. The open U is thus the locus where at least one f_i is nonvanishing. Thus we can write it as

$$U = \bigcup_{1 \leq i \leq r} X \setminus V(f_i) = \bigcup_{1 \leq i \leq r} D(f_i).$$

Thus, we are done.

■

Corollary 2.0.12. *The distinguished opens form a basis for the Zariski topology.*

Proposition 2.0.13. *We have*

$$\mathcal{O}_X(D(f)) = \left\{ \frac{g}{f^n} \mid g \in A(X) \right\} = A(X)[f^{\mathbb{N}}]^{-1}$$

for all $f \in A(X)$. We write this localization as $A(X)_f$. In particular, every function on $D(f)$ can be written as a ratio of two polynomials globally on $D(f)$. Hence $\mathcal{O}_X$ is quite tractable to compute as a ring localization.

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